



Design of Adsorption–Diffusion Dual-Driven MOF Membranes for Efficient CO₂ Separation

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Abstract: Metal–organic frameworks (MOFs) offer exceptional tunability in pore structure and functionality, holding considerable potential for advanced gas separation membranes. However, their advancement is restricted by the permeability–selectivity trade-off and inefficient empirical screening. Herein, we propose an adsorption–diffusion dual-driven design strategy balancing moderate CO₂ adsorption affinity ($1 < \alpha_{\text{ads,HTCS}} < 10$) and high CO₂ diffusivity selectivity ($\alpha_{\text{diff,HTCS}} > 10$) within a pore-limiting diameter range of 3.8–4.4 Å. This rationale is corroborated by high-throughput computational screening and experimental membrane performance. Three **yfm**-topology MOF membranes—CAU-10H, CAU-10pydc, and KMF-1 with finely tuned pore microenvironments were synthesized, with the former two meeting the proposed criteria and KMF-1 serving as a counterexample. As anticipated, CAU-10H and CAU-10pydc exceed the 2019 upper bound for CO₂/CH₄ separation, with CAU-10pydc exhibiting a remarkable CO₂ permeability of ~2847 Barrer and a selectivity of 185, outperforming most state-of-the-art membranes. Moreover, despite sharing the same kinetic diameter as CO₂, C₂H₂ exhibits 1.5 times higher adsorption affinity, resulting in a significantly lower permeability of only 39 Barrer under the same conditions. These results confirm the adsorption–diffusion dual-driven design principle, providing both theoretical insight and practical guidance for other challenging gas separation scenarios.

Introduction

Separation processes are pivotal in a wide range of industrial applications, with carbon capture receiving increasing attention in the context of global warming.^[1] Currently, the most widely applied separation techniques—such as thermally driven distillation, absorption, and adsorption—are highly energy intensive, accounting for 45%–55% of global industrial energy consumption.^[2] In contrast, membrane-based separation technology, driven by chemical potential gradients across the membrane, eliminates the phase change

process and can reduce energy consumption by up to 90% compared with conventional methods.^[3] Among the diverse available membrane materials, including polymeric, inorganic, and mixed-matrix membranes (MMMs), metal–organic framework (MOF) membranes stand out because of their highly tunable structural characteristics, such as pore size, pore environment, and sub-nanometer-level framework geometry.^[4,5] Moreover, the vast diversity of metal nodes and organic ligands enables the design of MOFs with tailored properties for specific separation needs.^[6]

However, the trade-off relationship between permeability and selectivity remains a major challenge hindering the advancement of membrane technology.^[7] From the separation mechanism perspective, molecular sieving, thermodynamic adsorption, and kinetic diffusion each present distinct advantages and limitations. For example, MOF membranes such as KAUST-7,^[8] ZIF-8,^[9,10] Co-gallate,^[11,12] and MOF-801,^[13] which are designed on the basis of the molecular sieving principle, exhibit exceptionally high selectivity in complex separations, including hydrogen purification, carbon capture, olefin/paraffin separation, and isomeric separation. However, their precisely matched pore sizes inherently restrict the diffusion rates of molecules, leading to low permeability.^[14] Conversely, membranes operating via the thermodynamic mechanism allow both gases to enter the pores, with selectivity primarily governed by the host-guest interaction strength and feed conditions.^[15] For example, the MOF-5 membrane delivers excellent CO₂/CH₄ selectivity (≈ 328 at 81.5% CO₂ feed) owing to its strong CO₂ adsorption affinity but completely loses this advantage under low partial-pressure conditions (≈ 2 at 50% CO₂ feed).^[16]

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 Supporting Information section

Alternatively, under the kinetic control mechanism, both gases can permeate the pores, and the selectivity arises from differences in diffusion rates. This typically yields higher permeability but lower selectivity than thermodynamically controlled processes.^[17,18] To address this limitation, a dual-driven separation strategy that synergistically integrates thermodynamic control (moderate adsorption affinity) and kinetic selection (significant differences in diffusion rates) within a single MOF membrane has emerged as a promising solution, though this approach remains largely unexplored.

Identifying MOFs that can optimally balance adsorption and diffusion properties for specific separation scenarios, such as natural gas purification (CO_2/CH_4), remains challenging due to the enormous structural diversity of MOFs.^[19] The traditional experimental approach relies on trial-and-error and can eventually achieve the desired separation performance, but this process is extremely time-consuming.^[20] High-throughput computational screening (HTCS) has emerged as a powerful alternative, enabling systematic and efficient evaluation of thousands of MOF candidates. Currently, the majority of HTCS studies have focused on discovering efficient adsorbents,^[21,22] a few have explored MOF membranes by evaluating adsorption selectivity ($\alpha_{\text{ads,HTCS}}$) using grand canonical Monte Carlo (GCMC) and diffusion selectivity ($\alpha_{\text{diff,HTCS}}$) via molecular dynamics (MD) simulations.^[23,24] Since different HTCS protocols employ varying screening metrics, databases, and operating conditions, the reported separation performance values cannot be directly compared on an absolute basis. Nevertheless, within each individual HTCS scheme, a common practice in these HTCS studies is to identify candidate MOF membranes by simultaneously maximizing both $\alpha_{\text{ads,HTCS}}$ and $\alpha_{\text{diff,HTCS}}$ that aiming for a theoretical high separation performance of MOF membranes. However, the fact that the extremely high α_{ads} in actual separation applications generally leads to a restricted diffusion rate and thus compromises MOF membrane separation performance has been largely overlooked in current HTCS studies.^[25,26] To address this gap, we employed HTCS to assist and accelerate the identification of optimal MOF membrane materials for CO_2 separation. Our strategy is based on a dual-driven separation mode that synergistically combines thermodynamic factor (moderate CO_2 adsorption affinity to avoid excessively strong host-guest interactions and kinetic factor (preferential CO_2 diffusion with large difference in diffusion rates), aiming to overcome the intrinsic permeability-selectivity trade-off.

Herein, we applied an HTCS protocol to more than 12000 MOF structures from the CoRE-2019 database, after removing duplicates and reconstructing incomplete frameworks.^[27] The overall research approach is illustrated in Figure 1. The HTCS screening process was used to efficiently evaluate a vast number of MOFs that fall into the designed range of adsorption selectivity ($\alpha_{\text{ads,HTCS}}$), diffusion selectivity ($\alpha_{\text{diff,HTCS}}$), proper pore-limiting diameters (PLD), and the overall membrane selectivity ($\alpha_{\text{mem,HTCS}}$) that is conducive to efficient CO_2/CH_4 separation (Figure 1 left). Guided by this adsorption–diffusion dual-driven principle, **yfm**-topology Al-MOFs based on CAU-10H was screened out. To further validate how a balanced interplay between adsorption and

diffusion can break the permeability–selectivity trade-off, two other Al-MOF membranes (CAU-10pydc and KMF-1) with identical topologies but sub-angstrom-scale pore size variations that fall within and outside the optimal PLD range, adsorption selectivity and diffusion selectivity ranges respectively were additionally selected for experimental membrane synthesis (Figure 1 right). As expected, CAU-10pydc MOF membranes surpassed the 2019 upper bound for CO_2/CH_4 mixed-gas separation relative to the counterpart membranes and outperformed most of the reported state-of-the-art membranes. The designed MOF membranes within the screened criteria strongly validate our dual-driven design principle and provide valuable insights for a wide range of separation applications.

Results and Discussion

High-Throughput Computational Screening for MOF Candidates

Metal–organic framework (MOF) membranes with both high permeability (P) and selectivity (α_{mem}) are generally identified as the optimal materials for gas separation. We performed HTCS to evaluate the CO_2 and CH_4 permeation behavior across a wide range of MOFs in the CoRE-MOFs 2019 database at 298 K under the prerequisite of their rigidity. This modification overcomes the current limitation of HTCS, which is unable to handle flexible or breathing MOFs. Structure–performance correlations were systematically analyzed to uncover the key descriptors governing separation performance. The influence of adsorption and diffusion selectivity ($\alpha_{\text{ads,HTCS}}$ and $\alpha_{\text{diff,HTCS}}$) on the overall CO_2/CH_4 separation performance was first examined. Approximately 99.8% of the MOFs (8780 structures) exhibited preferential CO_2 adsorption (Figure S1, $\alpha_{\text{ads,HTCS}} > 1$), indicating that the majority of MOFs tend toward thermodynamic-controlled separation for CO_2 , which can be attributed to their larger quadrupole moment and stronger Lewis acid–base interactions with MOFs than those of CH_4 .^[28] However, among these CO_2 adsorption-selective MOFs, only 1581 structures demonstrated CO_2 diffusion selectivity ($\alpha_{\text{diff,HTCS}} > 1$) related to kinetically-controlled separation for CO_2 (data points above the dashed line in Figure S2). Within the simultaneously adsorption and diffusion-selective MOFs ($\alpha_{\text{ads,HTCS}} > 1$, $\alpha_{\text{diff,HTCS}} > 1$, 1581 MOFs), a volcano-shaped relationship was observed between the diffusion coefficient (D_{CO_2}) and the adsorption selectivity ($\alpha_{\text{ads,HTCS}}$), as shown in Figure 2a. MOFs with moderate adsorption strengths ($\alpha_{\text{ads}} < 10$) exhibited the highest D_{CO_2} values (2–4 cm^2/s), indicating low diffusion resistance. In contrast, those with stronger adsorption ($\alpha_{\text{ads,HTCS}} > 10$) showed significantly suppressed diffusion, as evidenced by the extremely low D_{CO_2} values ($< 1 \text{ cm}^2/\text{s}$). These results suggest that an optimal balance between adsorption affinity and molecular mobility is essential for efficient diffusion. Therefore, a moderate CO_2 adsorption selectivity range within the range of $1 < \alpha_{\text{ads,HTCS}} < 10$, was identified as the most preferable condition. Besides, to achieve the adsorption–diffusion dual-driven membrane design, the molecular sieving mechanism related

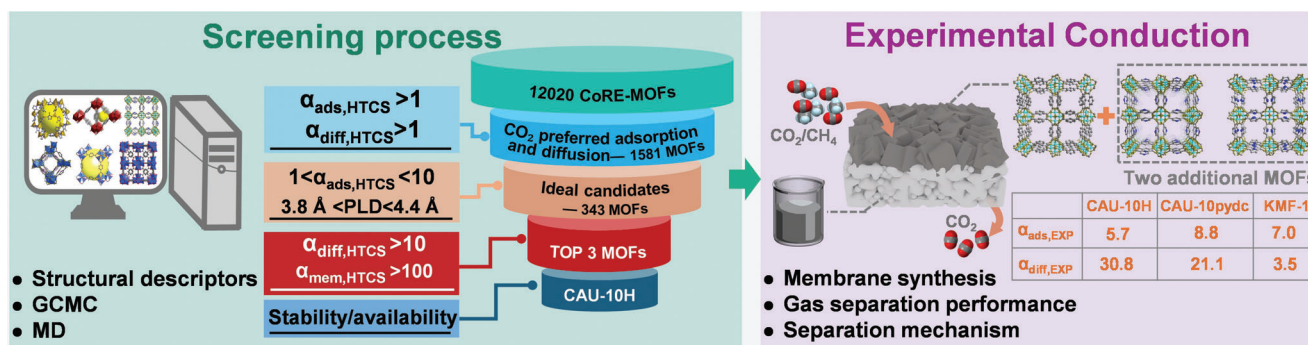


Figure 1. Workflow for the design of adsorption–diffusion dual-driven MOF membranes for CO₂ separation.

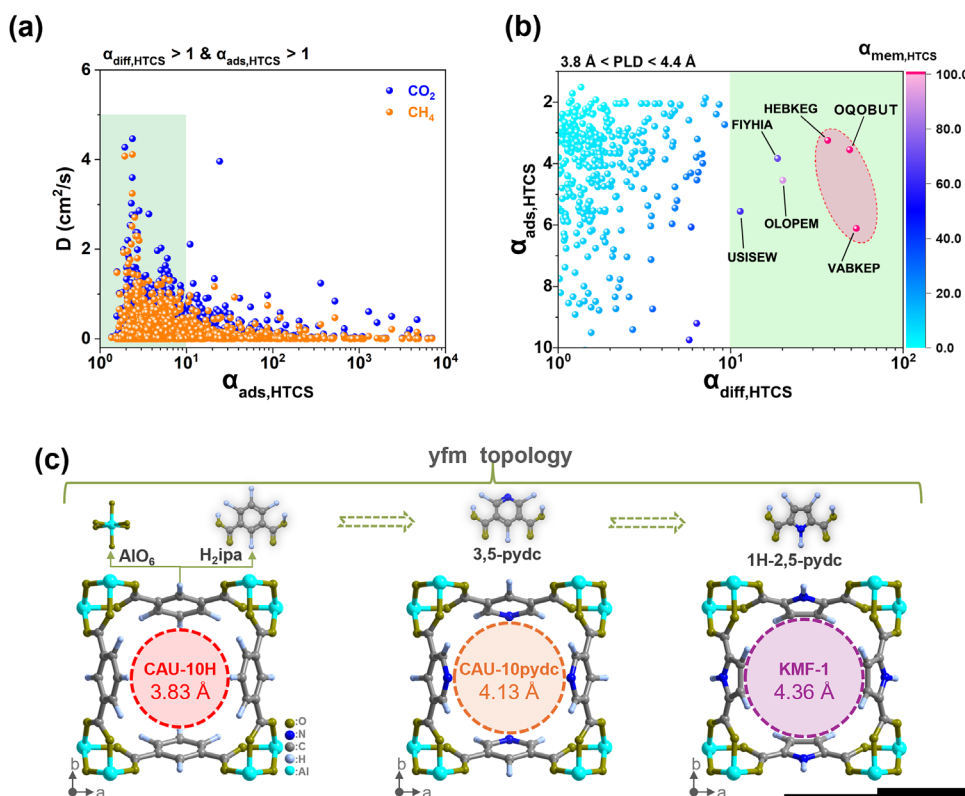


Figure 2. High-throughput computational screening highlights the importance of balancing adsorption and diffusion for achieving CO₂-selective transport. a) Diffusion coefficient of CO₂ (D_{CO_2}) and CH₄ (D_{CH_4}) for the 1581 structures that fall with $\alpha_{\text{diff,HTCS}} > 1$ and $\alpha_{\text{ads,HTCS}} > 1$ (the green-shaded region highlights cases where moderate adsorption selectivity with $\alpha_{\text{ads,HTCS}} < 10$ guarantee the higher CO₂ diffusion coefficient (D_{CO_2} : 2–4 cm²/s) and therefore the minimal diffusion resistance to both CO₂ and CH₄); b) Relationship between $1 < \alpha_{\text{ads,HTCS}} < 10$ and $\alpha_{\text{diff,HTCS}} > 1$ for CoRE-MOFs with PLDs ranging from 3.8 to 4.4 Å; c) Crystal structure of OQOBUT and two other crystals obtained by replacing original ligands.

MOF structures were also excluded by selecting a pore size range of 3.8–4.4 Å (Figures S3b), which allows the entrance of both gases. This movement further narrows the number of MOF structures to 343. Furthermore, a higher diffusion selectivity usually contributes to the desired membrane selectivity ($\alpha_{\text{mem,HTCS}} = \alpha_{\text{ads,HTCS}} \times \alpha_{\text{diff,HTCS}}$), hence, an efficient diffusion selectivity of $\alpha_{\text{diff,HTCS}} > 10$ (highlighted in green frame in Figure 2b) with obvious advantage over the rest of MOF structures was selected. Among the screened MOFs, three structures—HEBKEG,^[29] VABKEP,^[30] and OQOBUT^[31] highlighted in the red frame (Figure 2b,

Figure S4)—stand out owing to their distinctly high CO₂/CH₄ diffusion selectivity with favorable HTCS-predicted membrane selectivity ($\alpha_{\text{mem,HTCS}} > 100$). Meanwhile, the other three MOFs, namely FITHIA,^[32] OLOPEM,^[33] and USISEW^[34] although with moderate $\alpha_{\text{ads,HTCS}}$, were excluded at this stage due to the relatively low $\alpha_{\text{diff,HTCS}}$.

Among the three selected MOF candidates from Figure 2b, OQOBUT (denoted here as CAU-10H)—despite its strictly 1D channels—was picked for experimental validation based on the following reasons: i) The HTCS-predicted the $\alpha_{\text{ads,HTCS}}$ of 3.6 and $\alpha_{\text{diff,HTCS}}$ of 48.9 (Figure S4)

was aligned with the rule of the adsorption–diffusion dual-driven design and within the selection range of the $1 < \alpha_{\text{ads,HTCS}} < 10$ and $\alpha_{\text{diff,HTCS}} > 10$. ii) Robust Al^{3+} -carboxylate coordination imparts 1D-channel with high structural stability and relies on green and inexpensive ligands, which is advantageous for future scale-up. In contrast, the alternative 1D-channel candidate HEBKEG^[29] (Figure 2b) involves the heavy metal of Cr^{3+} center and HCl during the synthesis and the 3D-channel candidate VABKEP^[30] (Figure 2b) is built from costly Ag^+ centers and butanedinitrile ligands. iii) The periodic trans-1D Al chain coordinated with isophthalic acid (H_2ipa) forms an alternating narrow-window ($3.83 \times 3.83 \text{ \AA}^2$) and wide-cavity ($6.02 \times 6.02 \text{ \AA}^2$) channel along the c-axis with **yfm** topology (Figures 2c and S5),^[35] and ligand engineering allows for identical **yfm** topology design of CAU-10pydc (PLD = $4.13 \text{ \AA}$) and KMF-1 (PLD = $4.36 \text{ \AA}$), constructed from 3,5-Pyridinedicarboxylic acid (3,5-pydc),^[36] and 1H-pyrrole-2,5-dicarboxylic acid (1-H-2,5-pydc),^[37] respectively, serving as experimental validation material (Figure 2c).

Further experimental adsorption measurements confirmed that CAU-10H exhibited a CO_2 uptake of $37.0 \text{ cm}^3/\text{g}$ at 298 K and 1 bar and an $\alpha_{\text{ads,EXP}}$ of approximately 5.7 obtained from the experimental isotherms using ideal adsorbed solution theory (IAST) selectivity (Figure S6), as well as an effective $\alpha_{\text{diff,EXP}}$ of 30.8, back-calculated from $\alpha_{\text{mem,EXP}}$ according to Equation (S-5). In addition to CAU-10H, CAU-10pydc and KMF-1 were also examined both by HTCS (Figure S7) and the experimental validation. As expected, the intermediate PLD of CAU-10pydc following the order of $\text{CAU-10H} < \text{CAU-10pydc} < \text{KMF-1}$ possessed the balanced adsorption selectivity and diffusivity selectivity through the perspectives from both the HTCS prediction ($\alpha_{\text{ads,HTCS}}$: $3.6 < 4.3 \approx 4.4$, $\alpha_{\text{diff,HTCS}}$: $48.9 > 29.2 > 2.6$) and the experiment validation ($\alpha_{\text{ads,EXP}}$: $5.7 < 8.8 \approx 7.0$, $\alpha_{\text{diff,EXP}}$: $30.8 > 21.1 > 3.5$, Figure S6, Equation S-5). The consistent overall trend for the results between HTCS and experimentation highlights the HTCS can be an efficient rough tool for the target MOFs screening, despite some minor discrepancies for the specific value. The reported MOFs KUAST-7,^[8] ZIF-8,^[38] and MOF-5^[16] (Table S1) and the feed ratio effect on MOF-5^[16] membrane separation performance (Table S2) were also evaluated to further validate the balanced adsorption–diffusion selectivity screening principle.

Synthesis and Performance Validation of Al-MOF Membranes

Polycrystalline Al-MOFs (CAU-10H, CAU-10pydc, and KMF-1) were fabricated through a precisely controlled hydrothermal synthesis strategy, and the corresponding CO_2 permeation measurements were conducted via the Wicke–Kallenbach approach (Figures S8, S9). The key synthesis parameters for the membranes were thoroughly optimized through the synthesis temperature and time, as demonstrated in the (Figures S10–S30). Scanning electron microscopy (SEM) images (Figures 3a–c) revealed a compact structure without visible cracks or intergranular defects for these

three membranes with membrane thicknesses within $10 \text{ }\mu\text{m}$. The 3D surface topography images obtained via atomic force microscopy (AFM) (Figure S28) closely matched the morphology shown in the SEM images. The 1D pore channels aligned along the (020) crystallographic direction (view along the b-axis) with an interplanar spacing of 1.085 nm were detected via high-resolution transmission electron microscopy (HRTEM, Figure 3d). The resulting CAU-10H, CAU-10pydc, and KMF-1 membranes exhibited sharp crystalline X-ray diffraction (XRD) reflections (Figure 3e), and with relative intensities comparable to those of the simulated patterns, confirming the successful formation of all three Al-MOF phases on the ceramic supports and their random orientation (Figure S29). As expected, the size of the pore channels (Figure S31) increased in the order $\text{CAU-10H} < \text{CAU-10pydc} < \text{KMF-1}$, with precise adjustments at the sub-angstrom level obtained from the 77 K nitrogen adsorption–desorption tests, which aligned well with the theoretically predicted trend of Zeo++ with pore sizes of 3.83, 4.13, and $4.36 \text{ \AA}$ (Figure S5).^[39]

The overall integrity of the as-synthesized Al-MOF membranes was evaluated through single gas permeation measurements (Figure 3f, Table S3). The gas permeation of H_2 , CO_2 , N_2 , CH_4 , C_2H_4 , C_3H_6 , C_2H_6 , and C_3H_8 basically follows the trend of decreasing gas permeability with increasing kinetic diameter for the above molecules and with a clear cutoff between CO_2 and N_2 (highlighted in the green frame). Moreover, the larger pore sizes of the membranes ($\text{CAU-10H} < \text{CAU-10pydc} < \text{KMF-1}$) resulted in higher gas permeability, which was correlated with the typical behavior of pore size effect on membrane permeability. The ideal CO_2/CH_4 selectivity for the CAU-10H, CAU-10pydc, and KMF-1 membranes reached 59.1, 74.4, and 29.9, respectively, significantly exceeding the Knudsen diffusion selectivity (0.6) (Figure 3g), confirming the integrity of the membranes. Notably, the intermediate pore size of CAU-10pydc ($4.13 \text{ \AA}$) resulted in the highest $\alpha_{\text{mem,EXP}}$ among these three MOF membranes, and this performance was strengthened by the application of equimolar mixed gases (which increased from 74.4 to 185.2, Figure 3g). This is mainly because the transport behavior of CO_2 and CH_4 gas molecules in such a membrane is both thermodynamically and kinetically controlled, as indicated by the slightly reduced CO_2 permeability and enhanced CH_4 permeability with increasing feed temperature from 298 K to 323 K (Figure S43). Specifically, the pore microenvironment in CAU-10pydc, which is rich in nitrogen sites arising from the 3,5-pydc²⁻ ligand (Figure 2c), can provide additional adsorption affinity sites for CO_2 molecules compared with the original CAU-10H (which offers only AlO_6 -based sites), and this is confirmed by its significantly higher CO_2 uptake of $88.3 \text{ cm}^3/\text{g}$ versus $37.0 \text{ cm}^3/\text{g}$ for CAU-10H (Figure S6), highlighting the critical role of thermodynamic control. Moreover, the intermediate pore size ($4.13 \text{ \AA}$) with saturated adsorbed CO_2 molecules may form a narrowed pore channel that can still allow the small gas of CO_2 molecules to pass through but hinder the large gas of CH_4 . In contrast, the pore sizes of CAU-10H ($3.83 \text{ \AA}$) and KMF-1 ($4.36 \text{ \AA}$) after CO_2 adsorption may be too restrictive for CO_2 and too permissive for CH_4 diffusion, as evidenced

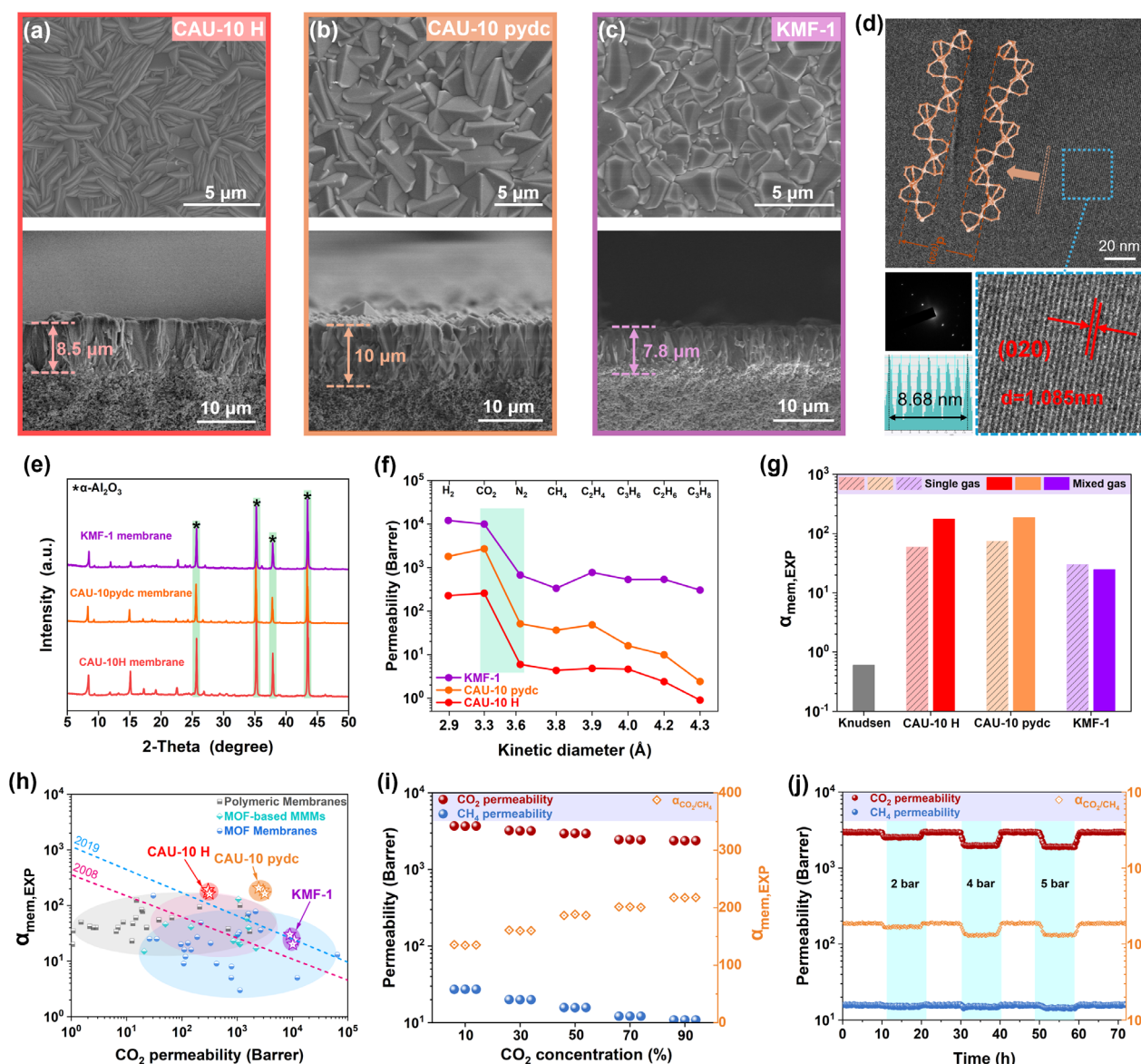


Figure 3. Membrane fabrication and gas separation performance evaluation. a–c) Top-view and side-view SEM images of CAU-10H, CAU-10pydc, and KMF-1 membranes, respectively. d) TEM image and schematic of the crystal channel structure of CAU-10pydc. e) XRD patterns of the CAU-10H, CAU-10pydc, and KMF-1 membranes. f) Single-gas permeability of the CAU-10H, CAU-10pydc, and KMF-1 membranes. g) Single and mixed gas selectivity for CO₂/CH₄ separation of CAU-10H, CAU-10pydc, and KMF-1 membranes. h) Separation performance comparison between the CAU-10H, CAU-10pydc, and KMF-1 membranes and other representative membranes reported in the literature. i) Separation performance of CAU-10pydc membranes for CO₂/CH₄ mixtures as a function of feed CO₂ concentration. j) Stability test of the CAU-10pydc membrane as a function of the operating pressure and time.

by the lowest P_{CO_2} (301.8 versus 2847.4 versus 9940.3 Barrer) for CAU-10H and the highest P_{CH_4} (1.7 versus 15.9 versus 417.3 Barrer, Table S4) for the KMF-1 sample, emphasizing the key function of kinetic control. Furthermore, with the CO₂ affinity site of the nitrogen heteroatom both in CAU-10pydc and KMF-1, the larger pore size of KMF-1 than CAU-10pydc showed a totally different gas permeation behavior which was with the decreased mixed gas selectivity (from 29.9 to 24.7, Figure 3g) relative to that of a single gas, further validating the crucial synergistic influence between thermodynamic and kinetic control. Consequently, the synthesized membranes with properties falling into the optimal PLD range (3.8–4.4

Å), adsorption selectivity (1–10), and diffusion selectivity (> 10) ranges (CAU-10H and CAU-10pydc versus KMF-1) all break the 2019 upper bound for the mixed gas CO₂/CH₄ separation (Figures 3h and S34, Table S11) and with CAU-10pydc sample possessing the immediate PLD (4.13 Å, Figure S5), the selectivity of adsorption (8.8, Figure S6) and diffusion (21.1) within the screened ranges exhibited the optimal separation performance and outperformed most of the reported membranes, including MOF-based pure and MMMs as well as polymeric membranes. This experimental behavior highlighted the importance of dual-function design for membrane materials in terms of both thermodynamic and

kinetic control. As a result, CAU-10pydc was selected as the optimal membrane sample for further study.

The effects of the CO₂ concentration and feed pressure on the optimal membrane separation performance and long-term stability were also examined. As the CO₂ feed concentration increased from 10% to 90% (Figure S3i and Table S9), P_{CO2} and P_{CH4} decreased because of the “narrowed” pore channels caused by Langmuir adsorption (logarithmic increase with pressure). A higher CO₂ feed concentration tends to create a narrower pore channel and corresponding stronger diffusion resistance for the larger gas of CH₄; hence, $\alpha_{\text{mem,EXP}}$ gradually increases. A similar trend was reported for other MOF and COF membranes.^[40–42] Meanwhile, the CAU-10pydc membrane retained outstanding separation performance even under low CO₂ feed conditions (1% to 5% partial pressure, Figure S39), achieving $\alpha_{\text{mem,EXP}}$ in the range of 258.0–325.3 (Table S5). In stark contrast, the $\alpha_{\text{mem,EXP}}$ of the MOF-5 membrane reported by Rui et al., governed mainly by thermodynamic control, nearly completely lost at a similar low CO₂ feed concentration.^[13] This comparison again confirmed the advantage of the synergistic effect between thermodynamic and kinetic control for membrane material design. For the pressure effect, both P_{CO2} and P_{CH4} slightly decreased (Figure S40 and Table S9) with increasing feed pressure from 1 bar to 5 bar. This is because gas permeability is defined as the transmembrane flux normalized by pressure and membrane thickness, whereas the actual diffusion flux follows Fick’s first law and is proportional to the concentration gradient of the dissolved gas molecules across the membrane (Figure S41).^[43] A higher pressure usually leads to the logarithmic (nonlinear) solubility behavior of the gas molecules and hence the resulting reduced gas permeability. Notably, the selectivity returned to its original value when the feed pressure was reduced to 1 bar within the 70 h testing period (Figure 3j), indicating an intact membrane structure under high-pressure separation conditions.^[44] It exhibited superior pressure resistance to the state-of-the-art zeolite membrane SSZ-13, with a smaller decline in CO₂/CH₄ selectivity (–27% versus –34%) when the feed pressure was increased to 4 bar.^[45] The exceptional pressure resistance and long-term stability of CAU-10pydc can be attributed to its robust framework, where strong covalent bonds exist between aluminum metal nodes (hard acids) and linkers (hard bases) on the basis of hard/soft acid-base theory.^[46] The retained crystallinity of both the membrane and the particles under harsh conditions (pH = 1–10, air exposure, water and hexane soaking) from the XRD patterns (Figures S35, S36), together with the intact MOF layer after ultrasonic treatment for 60 min (Figure S37) and the thermal stability up to 390 °C (Figure S38), further demonstrate the robust nature of the framework. Motivated by this stability, we further assessed the CO₂/CH₄ separation performance of the CAU-10pydc membrane under wet feed conditions (Figure S42). Upon switching from dry to wet feed, the P_{CO2} decreased from 2877.5 to 1792.1 Barrer, whereas P_{CH4} was only slightly reduced and thus $\alpha_{\text{mem,EXP}}$ decreased marginally to ~121. This suggests that water molecules compete with CO₂ for adsorption sites and partially occupy the transport pathways, thereby imposing additional resistance to CO₂ diffusion, consistent

with the trend reported by Fan et al.^[47] After the CAU-10pydc membrane was regenerated at 150 °C in a vacuum oven and tested again under dry gas, both P_{CO2} and $\alpha_{\text{mem,EXP}}$ almost returned to their initial values, confirming its excellent resistance to water vapor and long-term operational stability under realistic conditions. In addition, the screened robust CAU-10pydc membrane also exhibits excellent performance in CO₂/N₂ separation (Figure S33, S34, Table S10), further underscoring its potential as a highly promising candidate for CO₂ capture for a wide range of CO₂ feed concentrations, including flue gas purification (typically 10%–15% CO₂) and natural gas separation (commonly 5%–70% CO₂).^[48]

Separation Mechanism

The membrane design principle of balanced adsorption affinity and diffusion rates for efficient CO₂ separation was further validated both experimentally and theoretically through molecular dynamics simulations. For adsorption strength control, we selected C₂H₂ as a probe because it has the same kinetic diameter as CO₂ but has markedly different quadrupole moments (1.5 times greater for C₂H₂ than that for CO₂, Table S6), which means that the interaction between C₂H₂ and the MOF framework is stronger than that for the pair of CO₂–MOF frameworks.^[49] The traditional isosteric heat of adsorption measurement is not able to quantitatively assess the difference in interaction strength between CO₂ (or C₂H₂) and the CAU-10pydc framework (Figure S46). To overcome this limitation, inverse gas chromatography under infinite dilution conditions (ID-IGC) was applied to detect subtle differences in host–guest interactions. Under such conditions, the probe molecules C₂H₂, CO₂, and CH₄ can fully interact with the surface of the material, and their retention behavior in the MOF-packed column is solely governed by the strength of their interactions with the framework.^[50] As expected, the weakest interacting gas of CH₄ was the first to elute, with a retention time (T_N) of 3.2 min, followed by CO₂ and C₂H₂, with T_N of 18.8 min and 26.7 min, respectively at 55 °C (Figure 4a). The approximately 1.5 times slower elution time reflects the 1.5 times greater affinity of C₂H₂ toward the nitrogen-rich channel of CAU-10pydc and the corresponding diffusion resistance than that of CO₂. A similar trend was also found at an elevated elution temperature of 80 °C (Figure S47). Consistently, the varied interaction strengths were also reflected in their largely different single and mixed gas (CO₂/CH₄ and C₂H₂/CH₄) separation performances in Figure 4b and Table S7. Specifically, for single gas permeation (Figure 4b left), the P_{CO2} reached 2710.7 Barrer, while the P_{C2H2} was only 21.5 Barrer—just 0.79% of the CO₂ permeability. Notably, this value was even lower than that of CH₄ (36.5 Barrer), despite CH₄ having a larger kinetic diameter than C₂H₂. Similarly, under mixed-gas conditions (Figure 4b, right), the P_{CO2} in the CO₂/CH₄ system reached 2847.4 Barrer, while the P_{C2H2} in the C₂H₂/CH₄ system dramatically dropped at just 39.4 Barrer—a mere 1.3% of the CO₂ value. This was accompanied by a significant suppression of P_{CH4}, which decreased from 15.7 Barrer in the CO₂/CH₄ system to only 4.6 Barrer (29.3% of its original value) in the C₂H₂/CH₄ system.

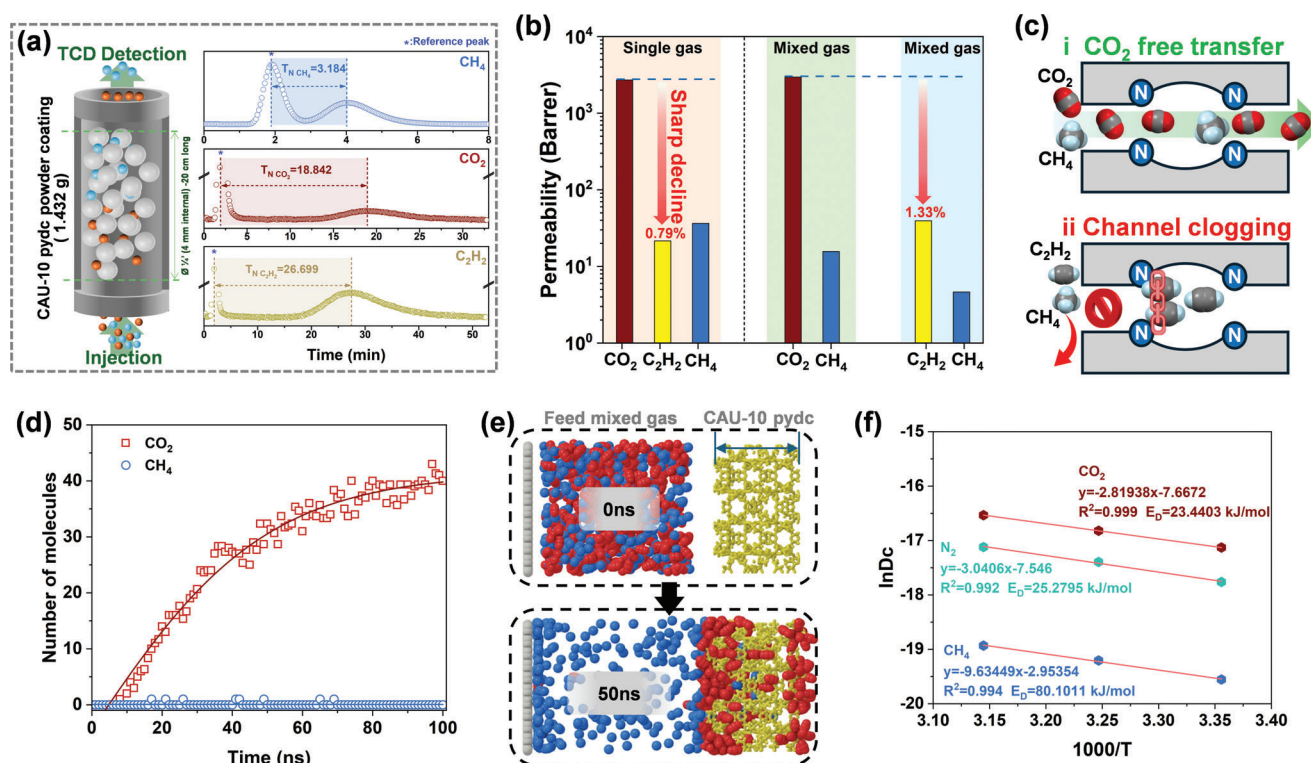


Figure 4. CO₂-selective transport in the CAU-10pydc membrane arises from a balanced adsorption–diffusion mechanism. a) Retention results from inverse gas chromatography at infinite dilution for three probe molecules with varying affinities, tested at 55 °C. b) Single and mixed gas permeation results for gas molecules with different affinities (C₂H₂, CO₂, and CH₄) through the CAU-10pydc membrane. c) Schematic illustration of the influence of nitrogen-rich channels on the transport of gas molecules with different affinities (CO₂ versus C₂H₂). d) Time-dependent variation in the number of CO₂ and CH₄ molecules transported through CAU-10pydc channels during molecular dynamics simulation. e) Molecular dynamics snapshots at 0 ns and 50 ns for CO₂ (red spheres) and CH₄ (blue spheres), respectively. f) Temperature-dependent diffusion coefficients of the CAU-10pydc membrane for CO₂ and CH₄.

This phenomenon reveals the adsorption–diffusion balance mechanism: CO₂ with moderate host–guest interaction can preferentially adsorb into the pores and diffuse rapidly along the pore walls (Figure 4c-i), imposing additional diffusion resistance on CH₄ (evidenced by the decrease in P_{CH_4} from single gas of 36.5 Barrer to the CO₂/CH₄ mixed gas of 15.7 Barrer, Figure 4b), whereas C₂H₂ with strong host–guest interaction induces pronounced channel blockage for both C₂H₂ and CH₄ (Figure 4c-ii), resulting in both greatly reduced permeability and poor C₂H₂/CH₄ selectivity ($\alpha_{mem,EXP} CO_2/CH_4 = 188$ versus $\alpha_{mem,EXP} C_2H_2/CH_4 = 8.5$). This result also underscores the rationale behind excluding MOFs with overly strong adsorption from the HTCS candidates (Figure 2a).

For the gas diffusion behavior, we performed equilibrium molecular dynamics simulations to visualize the permeation of CO₂ and CH₄ through the CAU-10pydc membrane (Table S12). CO₂ with a small kinetic diameter can be easily transported through the membrane quickly, whereas CH₄ with a large kinetic diameter is effectively prevented from entering the channels of the membrane, as shown in Figure 4d and snapshots of the CO₂ and CH₄ transport status from 0 ns to 50 ns (Figure 4e). A well-agreed phenomenon was also recorded for the CO₂/N₂ gas pair (Figure S48). Consistent with these simulation results, the

experimental results also demonstrate increasing diffusion activation energy (E_D) in the order of increasing gas kinetic diameter, with values of 23.4, 25.3, and 80.1 kJ/mol for CO₂ (3.3 Å), N₂ (3.64 Å), and CH₄ (3.8 Å), respectively as shown in Figure 4f (see, Figures S43, S44, and Table S8). Moreover, the time-resolved adsorption measurements further corroborated the fastest diffusion rate for CO₂ gas, followed by N₂ and CH₄ (Figure S45). Conclusively, on the basis of the above discussion, both the moderate adsorption affinity and fast diffusion rate for the targeted permeated gas species contributed to efficient gas pair separation, emphasizing the importance of the adsorption–diffusion dual-driven separation principle. Importantly, in addition to the disc membranes, high-quality tubular CAU-10pydc membranes with a length of 13 cm can also be successfully fabricated on commercially available α -alumina supports (Figure S49). An average P_{CO_2} of 2760 ± 360 Barrer and an average $\alpha_{mem,EXP}$ of 103 ± 17 (Figure S50) were obtained from the six tubular membranes prepared in the same batch, suggesting the feasibility of scaling up toward high packing density modules and their potential for CO₂ capture, despite the reduced gas separation performance relative to the disc membrane due to the lack of synthesis protocol optimization which will be the future work.

Conclusion

This study establishes an adsorption–diffusion dual-driven design principle for MOF membranes as a rational strategy to overcome the inherent permeability–selectivity trade-off in CO₂ separation. By simultaneously leveraging moderate CO₂ adsorption ($1 < \alpha_{\text{ads,HTCS}} < 10$) and preferential CO₂ diffusion ($\alpha_{\text{diff,HTCS}} > 10$), along with tailored pore size distributions (PLD = 3.8–4.4 Å) both from HTCS screening and actual experimental performance, this separation mode enables two out of three MOFs (CAU-10H, CAU-10pydc and KMF-1) that falling within these ranges break the 2019 upper bound for CO₂/CH₄ separation and with CAU-10pydc demonstrating the most optimal separation performance with CO₂ permeability (~2847 Barrer) and CO₂/CH₄ (185) outperforming most reported MOF membranes. Notably, it also demonstrated exceptional thermal, chemical, mechanical, and operational stability across a wide CO₂ concentration range (1%–90%) and up to 5 bar pressure, highlighting its potential for continuous industrial CO₂ capture. These findings underscore the effectiveness of the adsorption–diffusion dual-driven separation mechanism as a guiding principle for designing high-performance membrane materials, while also offering a general framework applicable to other targeted separation scenarios.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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